How to read research papers?

And not get mad.

Michał Włodarczyk, Vladimir Zaigrajew
March 2026

What is a research paper?

- A scientific report based on the current state-of-the-art of a specific domain, detailing novel findings or methods.
- A peer-reviewed contribution to human knowledge.
- A highly structured conversation between researchers across time and space.
- It's a building block you use to solve your own problems, back up your claims, or build your own narrative.

You probably already know all that.

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Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

1 Introduction

Recurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and

[*] Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

[†] Work performed while at Google Brain.

[‡] Work performed while at Google Research.

31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.

As an example we will look at
Attention is all you need
(Vaswani et al. 2017)
One of the most cited papers in ML ever.

What kind of research papers there are?

1. **Survey / Review Papers** - Prace przeglądowe
2. **Methodological Papers** - Prace metodologiczne
3. **Position Papers** - Prace opiniotwórcze
4. **Data Papers** - Prace o zbiorach danych
5. **Meta-Analyses** - Metaanalizy
6. **Theoretical Papers** - Prace teoretyczne
7. **Case Studies** - Studium przypadku)
8. **Short Papers / Letters** - Krótkie komunikaty
9. **Technical Reports** - Raporty techniczne
10. **Replication Studies** - Badania replikacyjne

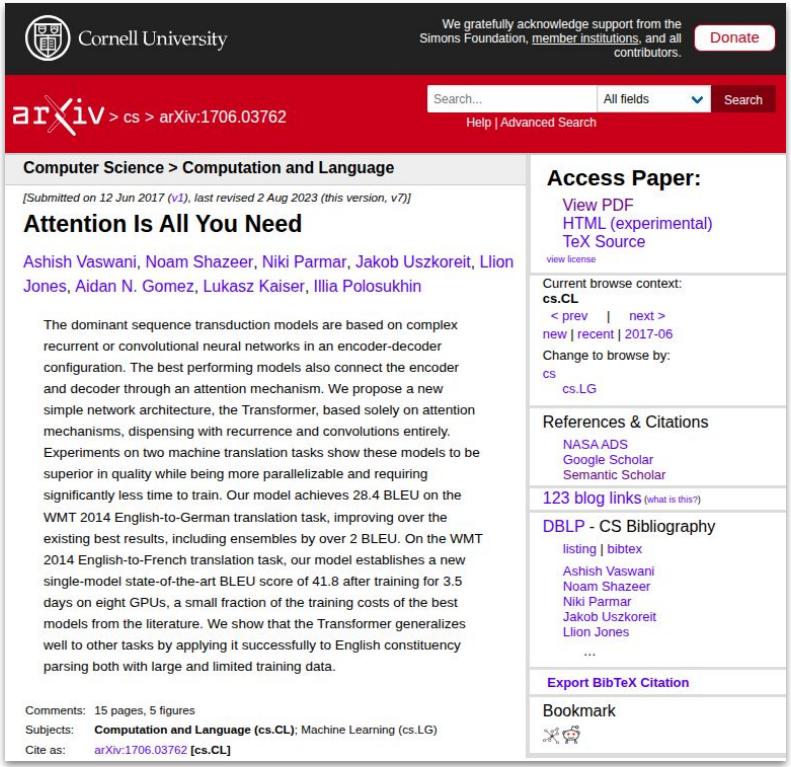
…and more

What makes papers stand out?

Published vs Unpublished

Papers may be:

1. **Abandoned** - Unfinished drafts or rejected manuscripts with no further action planned.
2. **Working Papers** - Active research in preparation or early drafts shared informally for feedback.
3. **Preprints** - Publicly accessible on open repositories (e.g., arXiv) but not yet peer-reviewed.
4. **Under Review** - Submitted to a journal or conference and actively undergoing evaluation or revision.
5. **Formally Published** - Successfully peer-reviewed, accepted, and officially released in a recognized venue.



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Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Comments: 15 pages, 5 figures

Subjects: **Computation and Language (cs.CL)**; Machine Learning (cs.LG)

Cite as: arXiv:1706.03762 **[cs.CL]**

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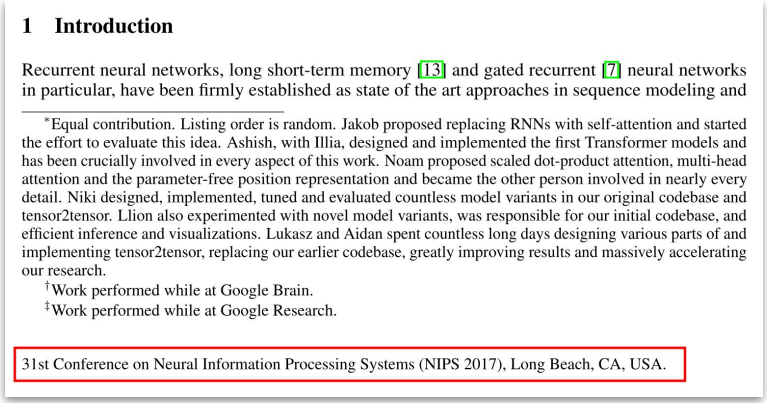
(n this case its the case 5.)

Note: We are talking here about how does it work in ML, it may vary in other fields

What makes papers stand out?

Publication Venue

- **Conferences vs. Journals** - In ML, top-tier conferences (e.g., NeurIPS) often match journal prestige but offer faster review cycles.
- **Selectivity Tiers** - Venue quality is gauged by acceptance rates and rankings such as CORE A* for conferences and Q1/Q2 for journals.
- **Peer-Review Baseline -** Publication confirms expert scrutiny of methodology and contribution, but venue tier determines how rigorous that bar was.
- **Impact Metrics -** Citation counts and h-index reflect influence, while CiteScore and Impact Factor measure venue reach.



1 Introduction

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Information about the venue in the paper

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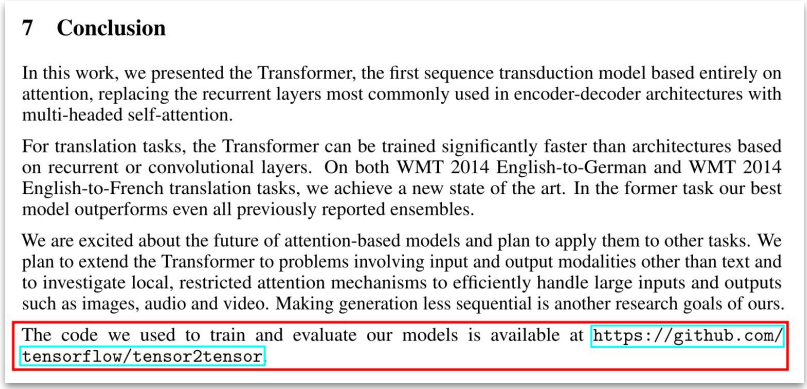
Is this paper worth my time?

A public code release signals practical value and author confidence:

- **Open Source Code** - Allows direct reuse, adaptation, and benchmarking.
- **Reproducibility** - Allows independent verification of results.
- **Models & Data** - Released weights or datasets lower the barrier to compare.
- **Maintenance** - Documentation and active responses signal long-term usability.
- **Benchmarks** - Some venues formally track reproducibility through required domain specific benchmarks.



7 Conclusion

In this work, we presented the Transformer, the first sequence transduction model based entirely on attention, replacing the recurrent layers most commonly used in encoder-decoder architectures with multi-headed self-attention.

For translation tasks, the Transformer can be trained significantly faster than architectures based on recurrent or convolutional layers. On both WMT 2014 English-to-German and WMT 2014 English-to-French translation tasks, we achieve a new state of the art. In the former task our best model outperforms even all previously reported ensembles.

We are excited about the future of attention-based models and plan to apply them to other tasks. We plan to extend the Transformer to problems involving input and output modalities other than text and to investigate local, restricted attention mechanisms to efficiently handle large inputs and outputs such as images, audio and video. Making generation less sequential is another research goals of ours.

The code we used to train and evaluate our models is available at https://github.com/tensorflow/tensor2tensor

Information about the codebase in the paper

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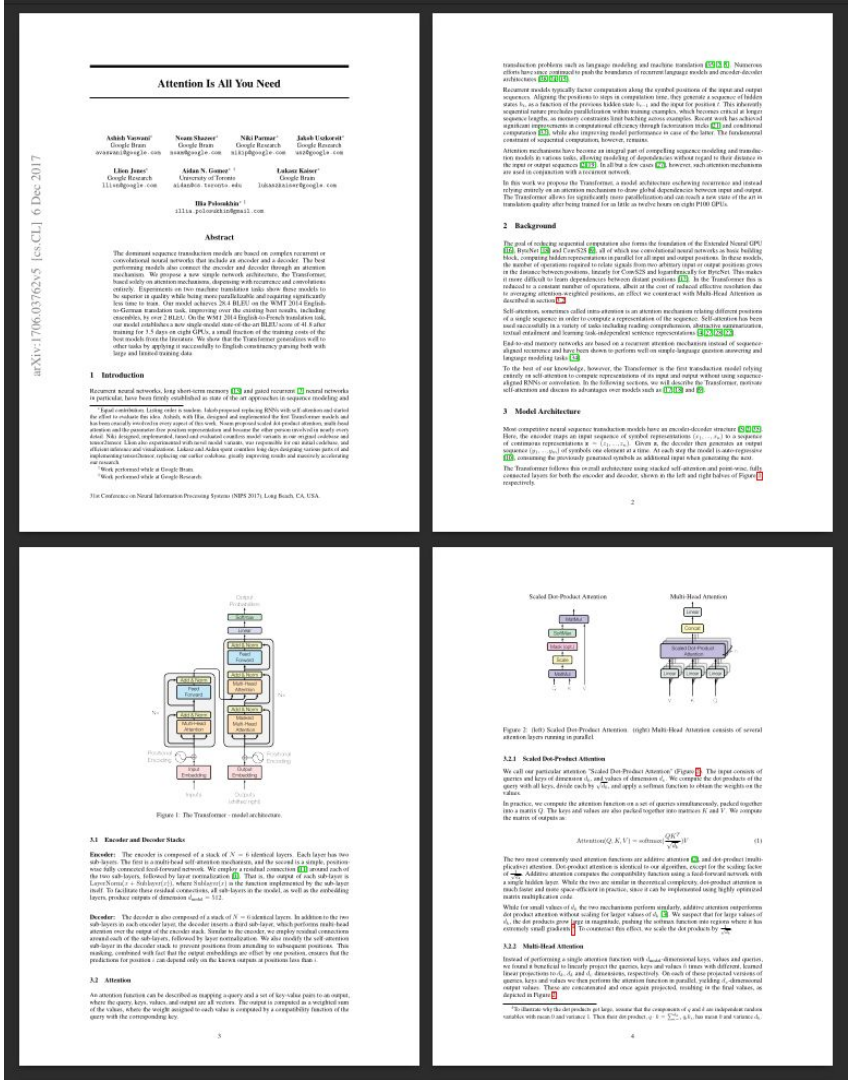
Anatomy of a Research Paper

A research paper typically follows a structured format, each section serving a distinct purpose:

- **Abstract** - High-level summary of the problem, method, and findings.
- **Introduction** - Motivation, problem statement, and claimed contribution.
- **Related Work** - Positioning the paper within existing literature.
- **Methodology** - The proposed approach, design choices, and technical novelty.
- **Experiments & Results** - Benchmarks, baselines, metrics, and key findings.
- **Conclusion** - Summary, limitations, and future directions.

…more info during lectures.



Where to Look When Skimming a Paper

Before committing to a full read, three sections give you the essential picture:

- **Abstract** - The condensed summary - problem, approach, and key result in a few sentences.
- **Last Paragraphs of the Introduction** - Authors typically summarize contributions and outline the paper structure here.
- **Conclusions & Future Work** - Restates what was shown and what remains unsolved.

Together these three sections let you assess relevance and merit in under five minutes.

transduction problems such as language modeling and machine translation [35, 2, 5]. Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [38, 24, 15].

Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t, as a function of the previous hidden state h_{t-1} and the input for position t. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [21] and conditional computation [32], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 19]. In all but a few cases [27], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

A showcase of what last paragraph of the introduction mentions in **Attention is all you need**

Paper Reading Strategies

You can approach a paper in many ways depending on your goal.

For Skimming:

1. Abstract
2. Introduction
3. Conclusion
4. Future Work
5. Methodology
6. Setup & Experiments
7. Related Works

As Written:

1. Abstract
2. Introduction
3. Related Works
4. Methodology
5. Setup & Experiments
6. Conclusion
7. Future Works

The skimming order prioritizes understanding the problem and prior work before evaluating the technical approach. Requires domain knowledge.

Paper Collection Tools

ResearchRabbit.AI is a graph based paper visualization tool that allows you to discover papers and see the context their in.

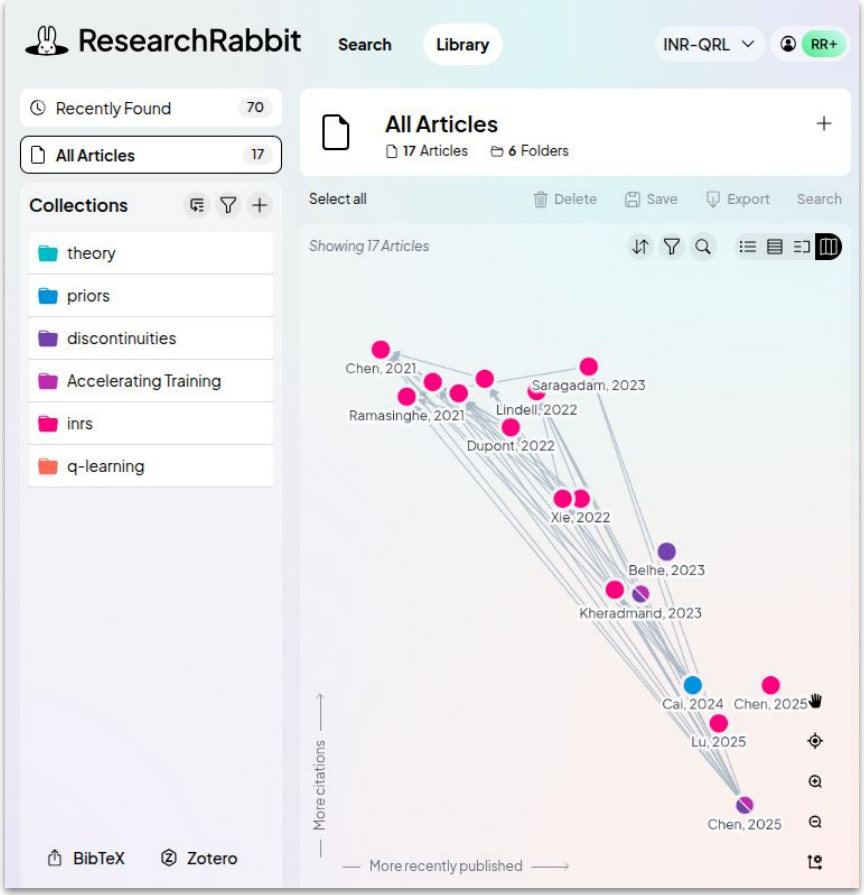
Collections - Organize papers into named folders by topic or project.

Citation Graph - Visualizes relationships between papers. Useful for tracing how ideas evolve over time.

Discovery - Recommends related papers based on your collection, helping map a research landscape quickly.

Export - Integrates directly with reference managers like BibTeX for seamless integration with paper writing tools like Overleaf.

This tool is primarily used during related works paper gathering.



https://app.researchrabbit.ai/

Paper Collection Tools

NotebookLM grounds an LLM assistant directly in your own uploaded documents.

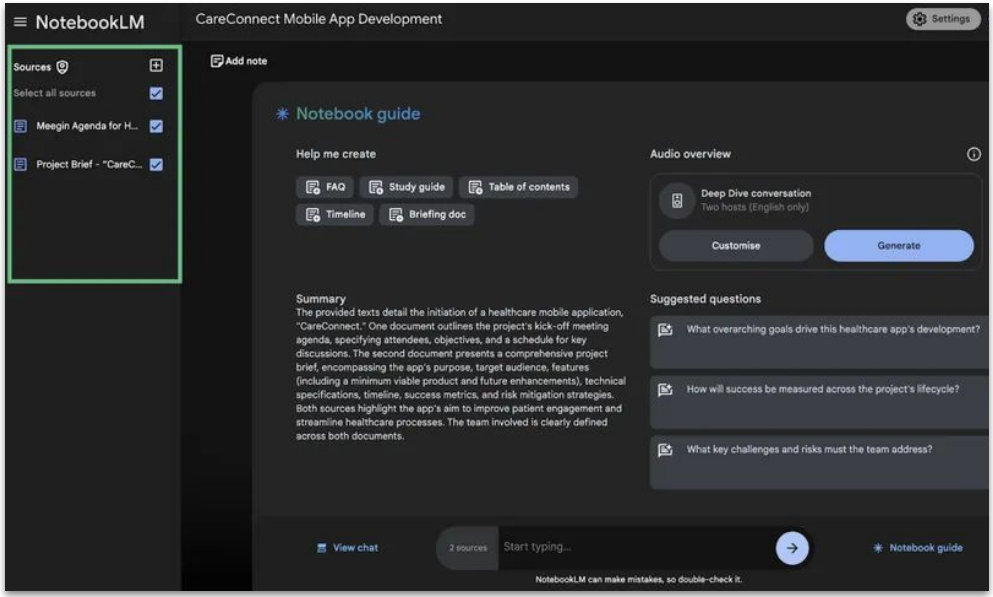
Source-Grounded Answers - Responses are based strictly on your uploaded papers, reducing hallucination risk.

Multi-Document Synthesis - Query across several papers at once to find connections and contradictions.

Notebook Guide - Auto-generates FAQs, study guides, and summaries from your sources.

Audio Overview - Generates a conversational deep-dive between two AI hosts — useful for passive review.

Best used when you have a focused set of papers (for example from research rabbit) and want to extract and cross-reference information quickly to evaluate preconceptions.



Notebook LM - part of Gemini

Paper Collection Tools

Perplexity is an LLM-powered search engine useful for early-stage research exploration.

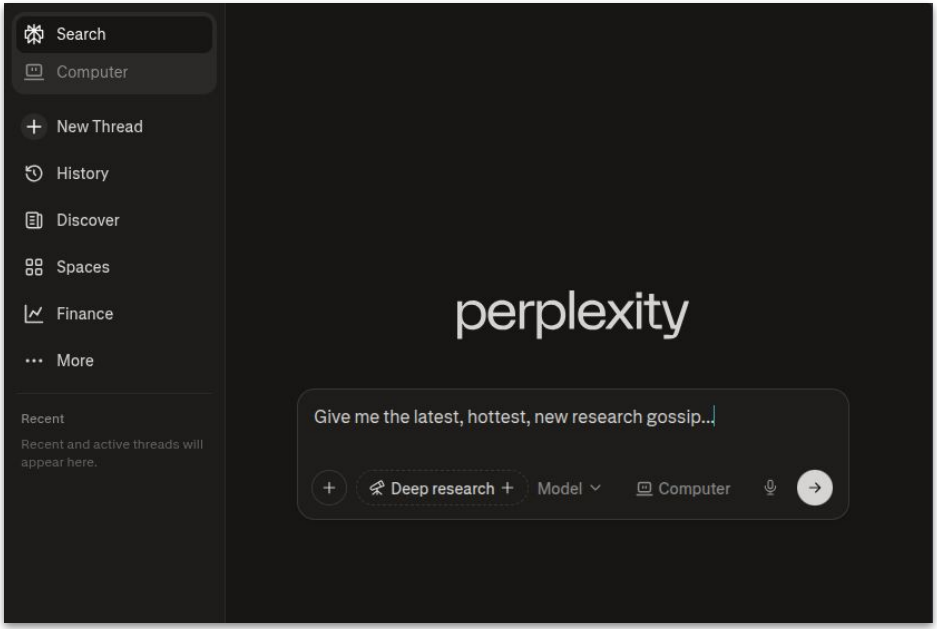
Natural Language Search - Query research topics conversationally instead of crafting keyword searches.

Deep Research Mode - Synthesizes information across multiple sources into a structured summary.

Discover - Surfaces trending and recent topics. WWW data gathering. General knowledge.

Spaces - Organize searches and threads by project or topic for ongoing research threads.

Best used for getting an initial lay of the land in an unfamiliar domain — **not a replacement for reading primary sources.**



https://www.perplexity.ai/

Other tools exist ofc.
If you find some that are useful please share them in Slack!

On Deep Research

AI assistants can accelerate literature review, but come with important caveats,

- **Hallucinations** - AI tools can confidently generate plausible but incorrect citations, authors, or findings. Always verify against the primary source.
- **Knowledge Cutoff** - Most models have a training cutoff and may be unaware of recent publications or developments.
- **Grounded vs. Ungrounded Tools** - Tools like NotebookLM work from your uploaded sources. Generally available chatbots do not.
- **Summarization Bias** - AI summaries may omit nuance, limitations, research gaps or miscommunicate the results present in the original paper.

Recommended use cases are rephrasing, summarizing known content, generating questions, and navigating unfamiliar terminology. **Treat AI outputs as a starting point, not a final answer**. The paper itself remains the ground truth. You’re responsible for what you write.

Please remember this during your work here and your thesis.

Now of you go!

I’ll send a list of papers and you choose one you’ll reproduce an experiment from and present it in 2-minute papers session.

Remember to send the experiment proposals.

If you find something interesting or find a better way to do so.

Let us know!